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Campus de Quixadá
Cálculo Diferencial e Integral I
Professor Ricardo Reis
Avaliação II — 2025.2

→ Usar dist de dois pontos

1. Seja a função $f(x) = -x^2 - x + 2$. Determine:

   a) (1.5) Os pontos da curva de $f(x)$ que possuem distância mínima até a origem.

   b) (0.5) O valor dessa distância mínima.

2. Um foguete parte da base com velocidade constante V. Uma antena localizada a uma distância fixa L da base de lançamento gira rastreando o foguete à medida que ele sobe. Quando a altura do foguete for H, determine:

   a) (2.0) A velocidade de afastamento entre o foguete e a antena.

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   b) (2.0) A velocidade angular do rastreador no instante em que essa altura é alcançada.

   ↳ def tg

   $\frac{dL}{dt} = 0$?

3. Calcule as seguintes integrais:

   a) (2.0) $\int x \sin x \, dx$

   b) (2.0) $\int_0^1 \frac{\sqrt{x^2+1}}{x} \, dx$ → $I = \int \sec^3 / tg$?!

(Dica:) transforme a função integrada em uma função racional.

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(1°) $\cancel{f(x) = -x^2 - x + 2}$

$x_1 + x_2 = -2$

$x_1 \cdot x_2 = -1$

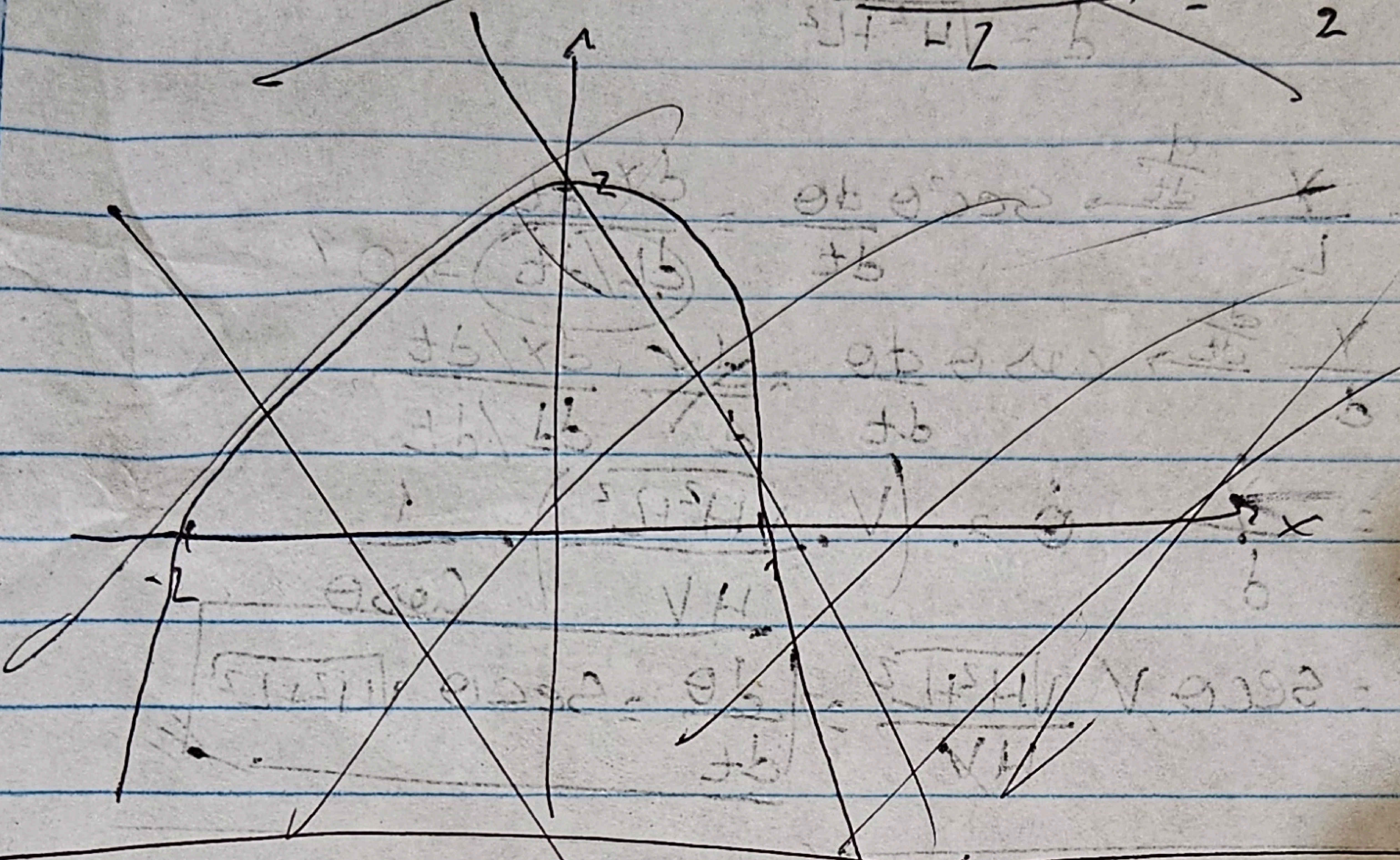
$\cancel{\Delta = 1 - 4.(-1).(2) = \Delta = 1}$

$\Delta = 1^2 - 4.(-1).2$

$\Delta = 1 + 8 = \Delta = 9$

$x = \dfrac{1 \pm \sqrt{9}}{-2} = \dfrac{1+3}{2}, -\dfrac{1-3}{2}$

$-2 \quad +1$



(10) $\cancel{y = -x_1^2 - x_1 + 2}$

$P = (0,0)$

$L^2 = x_1^2 + (-x_1^2 - x_1 + 2)^2$

$2L\dfrac{dL}{dx_1} = 2x_1 + 2[-x_1^2 - x_1 + 2].(-2x_1 - 1)$ $\quad \dfrac{dL}{dx} = 0$

$0 = 2x_1(+2x_1^2 - 2x_1 + 4)(-2x - 1)$

$0 = 2x_1$

$y_1 = x_1^2 - x_1 + 2$

$\dfrac{dy}{dx_1} = 2x_1 - 1$

$L^2 = x_1^2 + y_1^2$

$2L\dfrac{dL}{dx_1} = 2x_1 + 2y_1\dfrac{dy_1}{dx_1}$

$0 = 2x_1 + 2(x_1^2 - x_1 + 2)\,2x_1 - 1$

$x_1 = \dfrac{1+3}{-2} = -2$

$x_2 = \dfrac{1-3}{-2} = 1$

$\Delta = b^2 - 4ac \rightarrow \Delta = -1^2 - 4.(-1)2 = \Delta = 1 + 8 = 9$

9º) $f(x) = -x^2 - x + 2$

$$x_1 = \frac{1 + \sqrt{1 - 4 \cdot -1 \cdot 2}}{-2} = \frac{1+3}{-2} = -2$$

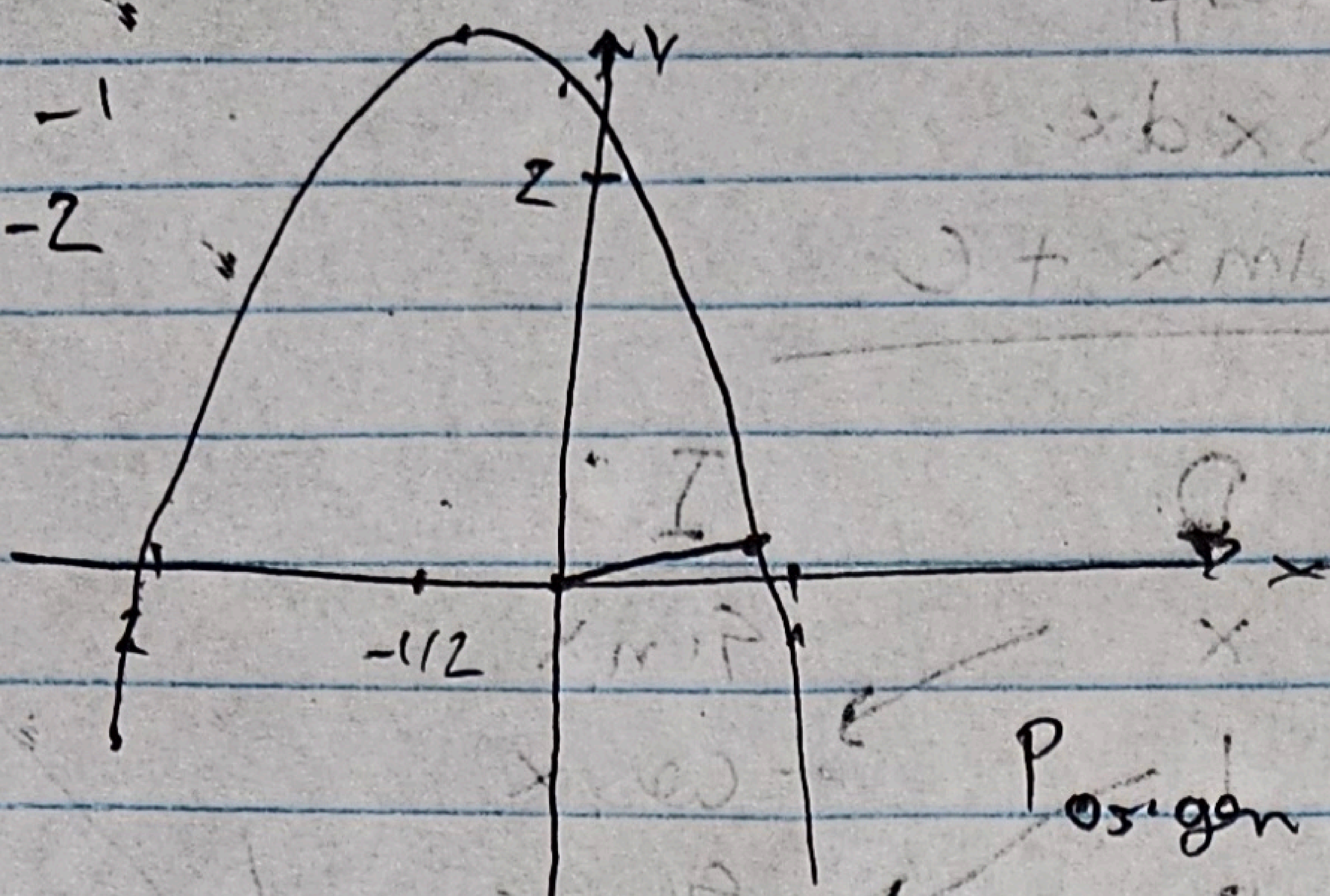
$$x_2 = \frac{1-3}{-2} = 1$$

$x_1 + x_2 = -1$

$x_1 \cdot x_2 = -2$

$x_1 = -2$

$x_1 = 1$



$-2x - 1 = 0$

$-2x = 1$

$x = -1/2$

$P_{origem} = (0;0)$

$L^2 = (x_1 - x_0)^2 + (y_1 - y_0)^2$ $\qquad y_1 = -x_1^2 - x + 2$

$L^2 = (x_1 - 0)^2 + (y_1 - 0)^2$

$L^2 = x_1^2 + (-x_1^2 - x_1 + 2)^2 \Rightarrow L = \sqrt{x_1^2 + (-x_1^2 - x_1 + 2)^2}$

$2L \frac{dL}{dx_1}$ $\qquad \frac{dL}{dx} = (x_1^2 + (-x_1^2 - x_1 + 2)^2)^{1/2}$

$$\frac{dL}{dx_1} = \frac{1}{2}\left[x_1^2 + (-x_1^2 - x_1 + 2)^2\right]^{-1/2} \cdot \left[2x_1 + 2(-x_1^2 - x_1 + 2) \cdot (-2x_1 - 1)\right]$$

$$\to 2L \frac{dL}{dx_1} = 2x_1 + \left[2(-x_1^2 - x_1 + 2) \cdot (-2x_1 - 1)\right]$$

$dL/dx_1 = 0$ $\to$ x^3 ???

$2L \cdot 0 = 2x_1 + \left[(-2x_1^2 - 2x_1 + 4) \cdot (-2x_1 - 1)\right]$

$0 = 2x_1 + [+4x_1^3 + 4x_1^2 - 8x_1 + 2x_1^2 + 2x_1 - 4$

$0 = 4x_1^3 + 6x_1^2 - 4x_1 - 4$

$0 = 2x_1^3 + 3x_1^2 - 2x_1 - 2$

$\frac{d}{}$ θ L y V $\boxed{d = ?}$ $\cancel{d^2 + y^2} = d^2 = L^2 + y^2 \to$ quando for H

2°) a) $\boxed{\dot{d} = \frac{HV}{\sqrt{H^2+L^2}}}$

$\dot{y} = V$

$y = H$

quando $y = H$

$d^2 = H^2 + L^2$

$d = \sqrt{H^2+L^2}$

$2d\frac{dd}{dt} = 2L\frac{dL}{dt} + 2y\frac{dy}{dt}$

$\frac{dL}{dt} = 0$

$d\dot{d} = \cancel{L\dot{L}} + y\dot{y}$

$\dot{d} = \frac{y\dot{y}}{d}$

b) $tg\theta = \frac{y}{L} \xrightarrow{\frac{d}{dt}} sec^2\theta\frac{d\theta}{dt} = \frac{dy/dt}{dL/dt} \to 0?$

$sin\theta = \frac{y}{d} \xrightarrow{\frac{d}{dt}} cos\theta\frac{d\theta}{dt} = \cancel{\frac{dy}{d}}\ \frac{dy/dt}{dd/dt}$

$cos\theta\,\dot{\theta} = \frac{\dot{y}}{\dot{d}}$ $\quad \dot{\theta} = \left(V \cdot \frac{\sqrt{H^2+L^2}}{HV}\right) \cdot \frac{1}{cos\theta}$

$\frac{d\theta}{dt} = sec\theta\,\cancel{V}\,\frac{\sqrt{H^2+L^2}}{H\cancel{V}} \Rightarrow \boxed{\frac{d\theta}{dt} = sec\theta\,\sqrt{H^2+L^2}}$

3°) b) $\int_0^1 \frac{\sqrt{x^2+1}}{x}dx = \int_0^1 \frac{\sqrt{x^2+1}}{x} \cdot \frac{\sqrt{x^2-1}}{\sqrt{x^2-1}}$

$\int_0^1 \frac{x^2-1}{x(\sqrt{x^2-1})} = \int_0^1 \frac{x^2}{x\sqrt{x^2-1}} - \int_0^1 \frac{1}{x\sqrt{x^2-1}}$

$\cancel{\int \frac{x}{\sqrt{x^2-1}}}$ $= \int_0^1 \frac{1}{x\sqrt{x^2-1}}$

$\frac{\sqrt{x^2+1}}{x}dx = \int_0^1 \frac{x^2+1}{x\sqrt{x^2+1}} = \int_0^1 \frac{x^2}{x\sqrt{x^2+1}}dx + \int_0^1 \frac{1}{x\sqrt{x^2+1}}dx$

$\frac{\sqrt{x^2+1}}{x}$

$b^2 = (a-b)(a+b)$

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3°) a) $\int x \sin x \, dx$; $u = x$ | $du = dx$; $v = -\cos x$ | $dv = \sin x \, dx$

$$\int x \sin x \, dx = -\cos x \cdot x + \int \cos x \, dx$$

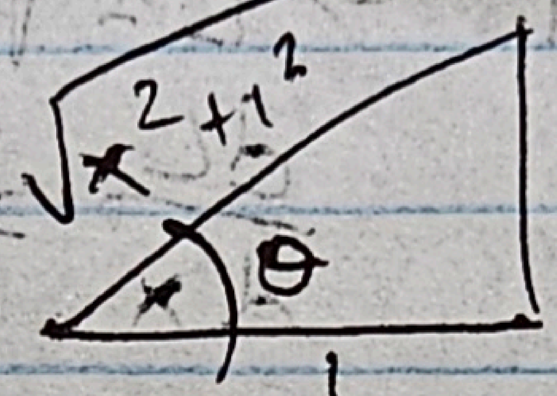
$$\int x \sin x \, dx = -\cos x \cdot x + \sin x + C$$

	D	I
+	x	$\sin x$
-	1	$-\cos x$
+	0	$-\sin x$

$$\int x \sin x \, dx = -\cos x \cdot x + \sin x + C$$

2,0

b) $\int_0^1 \frac{\sqrt{x^2+1}}{x} dx$



$tg\theta = \frac{x}{1}$ $\sec\theta = \sqrt{x}$

$\sec^2\theta \, d\theta = dx$

$x = tg\theta$; $x \Rightarrow 0$ $\theta \to$

$x \Rightarrow 1$ $\theta \to \pi$

$$\int_0^{\pi/4} \frac{\sqrt{tg^2\theta + 1} \cdot \sec^2\theta \, d\theta}{tg\theta}$$

$$\int_0^{\pi/4} \frac{\sqrt{\sec^2\theta}}{tg\theta} \sec^2\theta \, d\theta = \int_0^{\pi/4} \frac{\sec\theta}{tg\theta} \sec^2\theta \, d\theta$$

$$\cancel{\int_0^{\pi/4} \sec \frac{1}{\cos^3\theta} \cdot \frac{\cos\theta}{\sin\theta} = \int_0^{} c}$$

$$\int_0^{\pi/4} \frac{(1+tg\theta)^3}{tg\theta} d\theta = \int_0^{\pi/4} \frac{1 + 3tg\theta + 3tg^2\theta + tg^3\theta}{tg\theta} d\theta$$

$$= \int_0^{\pi/4} \frac{1}{tg\theta} d\theta + \int_0^{\pi/4} \frac{3tg\theta}{tg\theta} d\theta + \int_0^{\pi/4} \frac{3tg^2\theta}{tg\theta} d\theta + \int_0^{\pi/4} \frac{tg^3\theta}{tg\theta} d\theta$$

$$= \int_0^{\pi/4} \frac{1}{tg\theta} d\theta \; 3\theta \Big|_0^{\pi/4} + \int_0^{\pi/4} 3tg\theta \, d\theta + \int_0^{\pi/4} tg^2\theta \, d\theta$$

$$= 3\theta \Big|_0^{\pi/4} + \int_0^{\pi/4} cotg\theta \, d\theta + 3\int_0^{\pi/4} tg\theta \, d\theta + \int_0^{\pi/4} tg^2\theta \, d\theta$$

!!!

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$\frac{\sqrt{3}}{2}$ 1 $\sqrt{3}$ $\frac{180}{4} = \frac{160}{4} + \frac{20}{4} = 40 + 5 =$